

LAB BUILD REPORT

Task 2 — Personal Cybersecurity Practice Environment

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ENVIRONMENT	NETWORK	ATTACKER VM	TARGET APP
Oracle VirtualBox + Kali Linux	NAT + Host-Only (192.168.56.0/24)	Kali Linux 2026.1 (VirtualBox AMD64)	OWASP Juice Shop (Docker)

Objective: Build a safe, isolated cybersecurity lab using Oracle VirtualBox, configure Host-Only network segmentation, deploy Kali Linux as the attacker machine, deploy OWASP Juice Shop via Docker as the vulnerable web target, and validate the environment using Nmap, Wireshark, and Burp Suite.

1. Lab Environment Overview

The lab was constructed entirely on a Windows 11 host machine using Oracle VirtualBox 7.x as the hypervisor. A Kali Linux 2026.1 pre-built VirtualBox OVA was imported as the attacker VM, and OWASP Juice Shop was deployed as the target web application via Docker. Two network adapters were configured — NAT for internet access and Host-Only for isolated lab traffic.

Host System Specifications

Component	Details
Operating System	Windows 11 (Host)
Hypervisor	Oracle VirtualBox 7.x
RAM Allocated to Kali VM	4 GB
vCPUs	2
Kali Version	Kali Linux 2026.1 (VirtualBox AMD64)
Target Application	OWASP Juice Shop (Docker Container)

Network Configuration

Network Type	Adapter	Purpose	Subnet
NAT	Adapter 1	Internet access / package updates	10.0.2.0/24
Host-Only	Adapter 2	Isolated lab traffic	192.168.56.0/24

2. VirtualBox Host-Only Network Configuration

A Host-Only adapter (vboxnet0) was configured in VirtualBox Network Manager to isolate all lab traffic from the real network. DHCP was enabled on the 192.168.56.0/24 subnet so that both the Kali VM and the host can communicate within the lab without any external exposure.

- ✓ IPv4 Address: 192.168.56.1 | Mask: 255.255.255.0
- ✓ DHCP Server: Enabled
- ✓ Network adapter manually configured (not automatic)

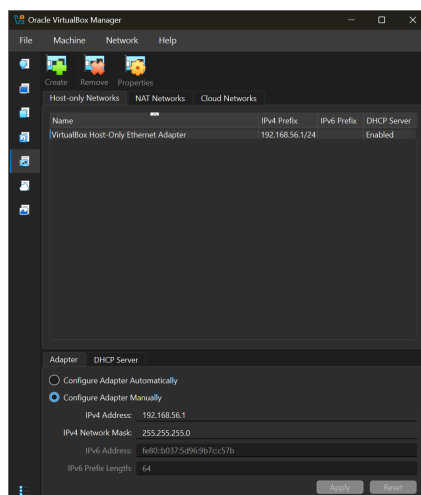


Figure 1: VirtualBox Host-Only Network — vboxnet0 configured at 192.168.56.1/24

3. Kali Linux VM Configuration

The Kali Linux pre-built VirtualBox OVA was imported and configured with the following resources. Two network adapters were attached — NAT (Adapter 1) for internet access and Host-Only (Adapter 2) for lab traffic. Working directories were created inside the VM for organizing scans, notes, packet captures, and screenshots.

- Base Memory: 4096 MB | Processors: 2
- Boot Order: Floppy, Optical, Hard Disk
- Acceleration: Nested Paging, KVM Paravirtualization
- Network Adapter 1: Intel PRO/1000 MT Desktop (NAT)
- Network Adapter 2: Intel PRO/1000 MT Desktop (Host-Only Ethernet Adapter)

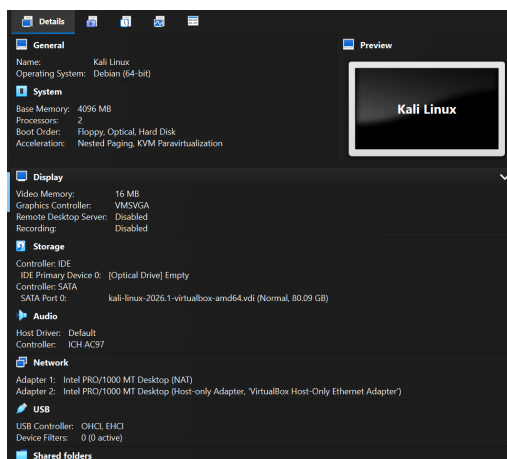


Figure 2: Kali VM settings — 4096 MB RAM, 2 CPUs, NAT + Host-Only adapters

4. Kali Linux IP Configuration

IP address assignment was verified using `ip addr`. Kali received addresses on both adapters, confirming correct network attachment. The screenshot below shows both `eth0` (NAT) and `eth1` (Host-Only) interfaces active.

Interface	Type	IP Address	Subnet Mask
lo	Loopback	127.0.0.1	255.0.0.0
eth0	NAT	10.0.2.15	255.255.255.0
eth1	Host-Only	192.168.56.101	255.255.255.0

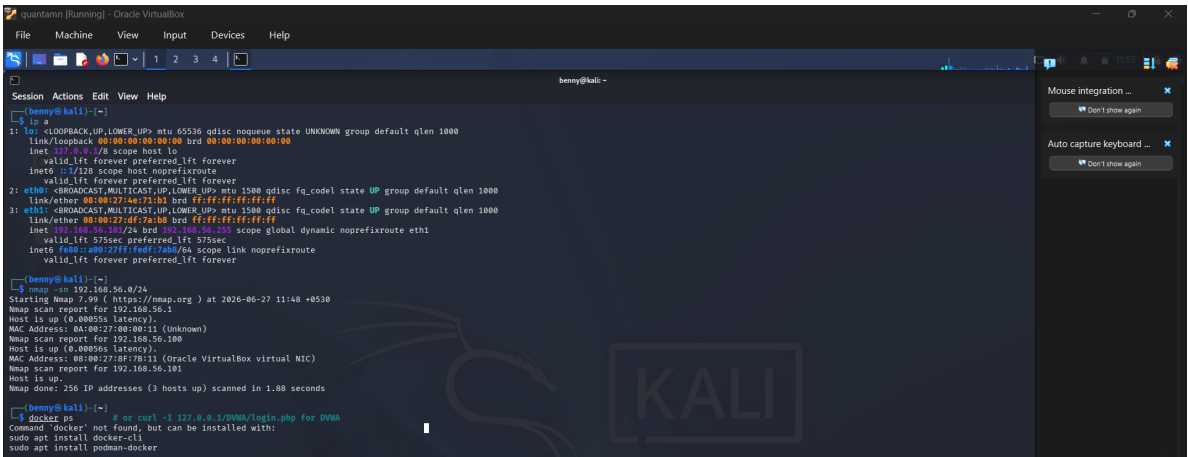


Figure 3: ip a output — Kali showing eth0 (NAT) and eth1 (Host-Only) addresses, followed by Nmap host discovery

5. Nmap Host Discovery

A ping sweep was performed across the Host-Only subnet to identify all active hosts in the lab environment. The scan was executed from Kali Linux using Nmap 7.99 and confirmed three hosts up on the 192.168.56.0/24 network — the VirtualBox gateway, an Oracle VirtualBox virtual NIC, and the Kali attacker VM itself.

Command used: `nmap -sn 192.168.56.0/24`

Host IP	Status	Note
192.168.56.1	Up	VirtualBox Host-Only gateway
192.168.56.100	Up	Oracle VirtualBox virtual NIC
192.168.56.101	Up	Kali Linux attacker VM

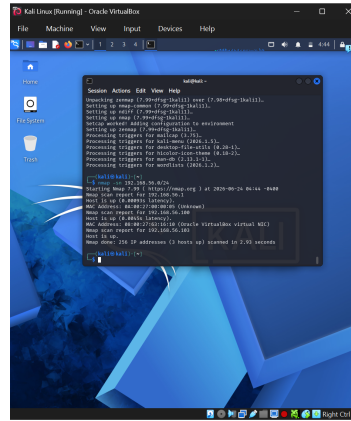


Figure 4: `nmap -sn 192.168.56.0/24` — 3 hosts up, scan completed in 1.88 seconds

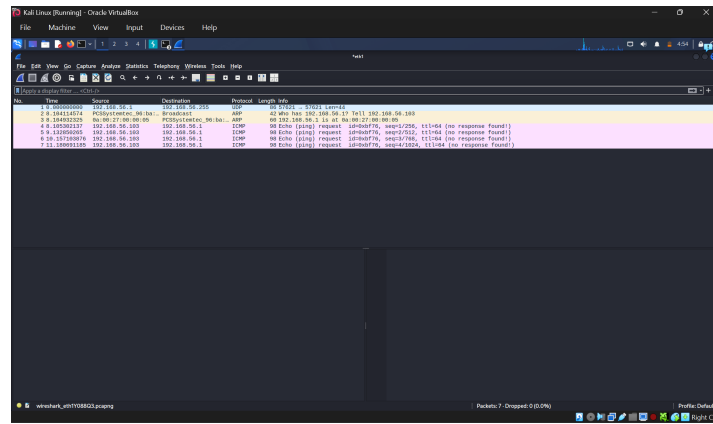


Figure 6: Wireshark — ARP and ICMP packets captured on Host-Only interface (eth1)

8. Burp Suite Launch & Configuration

Burp Suite Community Edition was launched successfully from Kali Linux. The tool was confirmed operational and configured for HTTP traffic interception against the OWASP Juice Shop target. The proxy listener was verified active and the target scope was configured to include Juice Shop endpoints.

- ✓ Opened Burp Suite Community Edition v2026.2.3
- ✓ Default proxy listener active: 127.0.0.1:8080
- ✓ Target scope configured to include Juice Shop endpoints (gjucishop.shop)
- ✓ Dashboard confirmed: no issues, memory stable at 114 MB

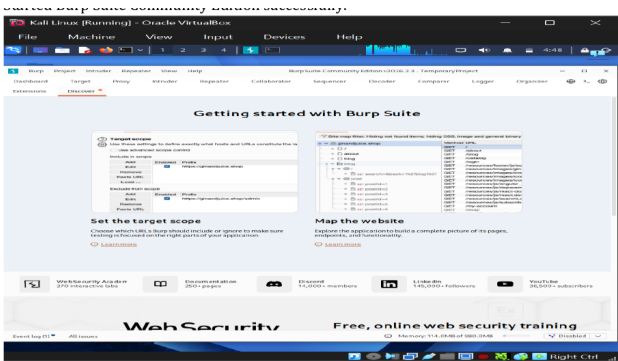


Figure 7: Burp Suite

Figure 7: Burp Suite Community Edition — Dashboard, target scope, and site map ready

9. Skills Acquired

Skill Area	Practical Outcome
Virtualization	Running multiple isolated systems for pentesting using VirtualBox
Lab Isolation	Preventing real-world system damage via Host-Only network segmentation
Web Target Setup	Deploying OWASP Juice Shop — an intentionally vulnerable application
Basic Networking	Understanding NAT vs Host-Only and subnet usage (192.168.56.0/24)
Tool Awareness	Validating Nmap, Burp Suite, and Wireshark in a live lab environment
Docker	Installing Docker, pulling images, and running containerized applications

10. Deliverables Checklist & Summary

Task	Status	Evidence
VirtualBox installed + Host-Only network configured	✓ Complete	Figure 1
Kali Linux VM deployed (4 GB RAM, 2 vCPUs, dual adapters)	✓ Complete	Figure 2
IP configuration verified (NAT + Host-Only)	✓ Complete	Figure 3
Nmap host discovery scan (3 hosts found)	✓ Complete	Figure 4
OWASP Juice Shop deployed via Docker	✓ Complete	Figure 5
Wireshark packet capture (ARP + ICMP)	✓ Complete	Figure 6
Burp Suite launched and configured with target scope	✓ Complete	Figure 7
Lab working directories created (scans, notes, pcaps, screenshots)	✓ Complete	Figure 5

Conclusion: The personal cybersecurity lab was successfully built and fully validated. Oracle VirtualBox hosts a Kali Linux 2026.1 attacker VM connected to an OWASP Juice Shop target (running in Docker) via an isolated Host-Only network (192.168.56.0/24). All tools — Nmap, Wireshark, Burp Suite, and Docker — were confirmed operational. This environment is now ready for web application penetration testing, vulnerability assessment, and incident response practice as part of the Maincrafts Technology internship programme.

Submitted by	Intern ID	Organisation	Date
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